

Lattice — Feature Showcase

Open this file with **Lattice** to see every feature rendered live.

Switch between **Edit**, **Preview**, and **Dual** view with the toolbar selector.

Try `Ctrl/Cmd+Shift+T` to auto-generate the Table of Contents below.

Table of Contents

- [Table of Contents](#)
- [Text Formatting](#)
- [Highlight Marks](#)
- [Lists](#)
 - [Unordered](#)
 - [Ordered](#)
 - [Definition-style \(using bold + indent\)](#)
- [Task Lists \(GFM\)](#)
- [Tables](#)
- [Code](#)
 - [Inline code](#)
 - [Fenced code blocks](#)
- [Blockquotes](#)
 - [GitHub-style Alerts \(rendered by GitHub and Lattice\)](#)
- [Mathematics — KaTeX](#)
 - [Inline math](#)
 - [Display math — Calculus](#)
 - [Display math — Linear Algebra](#)
 - [Display math — Series & Limits](#)
- [Mermaid Diagrams](#)
 - [Flowchart — Lattice Edit Cycle](#)
 - [Sequence Diagram — File Open Flow](#)
 - [Class Diagram — Core Data Model](#)
 - [Pie Chart — Supported Platforms](#)
- [Images](#)
- [Links](#)
- [HTML in Markdown](#)
- [Keyboard Shortcuts Reference](#)

Text Formatting

Normal paragraph text flows here. Lattice renders Markdown as **close to GitHub** as possible, so files look identical whether viewed in the editor preview, on GitHub, or printed.

Style	Syntax	Result
Bold	<code>**bold**</code>	bold
Italic	<code>*italic*</code>	<i>italic</i>
Bold + italic	<code>***both***</code>	<i>both</i>
Strikethrough	<code>~~strikethrough~~</code>	strikethrough
Inline code	<code>`code`</code>	<code>code</code>
Superscript	<code>X<sup>2</sup></code>	X^2
Subscript	<code>H<sub>2</sub>O</code>	H_2O

Note — sub/superscript: CommonMark/GFM has no native sub/superscript syntax. Lattice renders `<sup>` and `<sub>` HTML tags directly (GitHub and Obsidian do too). We recommend, though that you use alternatives that work everywhere: KaTeX math (`$X^2$` → X^2 , `$H_2O$` → H_2O) or Unicode characters (`x²` , `H₂O`). The linter flags these tags as a portability hint — they may not render in all renderers.

Highlight Marks

Highlight marks are a Lattice-exclusive feature. Wrap any text in `==double equals==` to render it with a **bright amber background**.

This works in **any context** — inside a sentence, `==across italic and bold spans==`, or on its own line:

This entire sentence is highlighted.

[!NOTE] Highlight marks are **not** rendered inside code blocks or inline code — `==this stays raw==` .

Lists

Unordered

- First item
- Second item
 - Nested item A
 - Nested item B
 - Deeply nested
- Third item with **bold** and *italic* text

Ordered

1. Install Lattice from the [releases page](#)
2. Open any `.md` file — double-click or via **File** → **Open**
3. Choose your view:
 - i. **Edit** — raw CodeMirror editor only
 - ii. **Preview** — rendered HTML only
 - iii. **Dual** — side-by-side (four layout variants)
4. Print with `Ctrl/Cmd+P` — always renders to a clean light theme

Definition-style (using bold + indent)

Local-first All files stay on your machine. No cloud sync required.

Documents as Code Plain-text `.md` files are diff-friendly, version-controlled, and AI-ready.

Task Lists (GFM)

GitHub Flavored Markdown task lists render as interactive checkboxes in Lattice.

- ☒ Install Lattice
- ☒ Open the demo file
- ☒ Explore Edit / Preview / Dual views
- ☐ Try `Ctrl/Cmd+Shift+T` to regenerate the Table of Contents
- ☐ Try `Ctrl/Cmd+Shift+L` to auto-align a pipe table
- ☐ Paste a screenshot with `Ctrl/Cmd+V` — it auto-saves to a sibling folder
- ☒ Open a second file in a new window (multi-window support)
- ☐ Try printing to PDF with `Ctrl/Cmd+P`

Tables

Lattice supports **GFM pipe tables** with left, center, and right alignment. Press `Ctrl/Cmd+Shift+L` with the cursor anywhere in the table to auto-format it.

Feature	Edit View	Preview View	Dual View
CodeMirror editor	✓	—	✓
Live rendered preview	—	✓	✓
Scroll sync	—	—	✓
Print (Ctrl+P)	✓	✓	✓
Highlight marks	✓	✓	✓
Mermaid diagrams	—	✓	✓
KaTeX mathematics	—	✓	✓

Code

Inline code

Use backticks for `inline code`. Great for `file/paths`, `commands`, and `variable_names`.

Fenced code blocks

Lattice passes code fences to the preview unchanged — syntax highlighting is provided by the renderer.

```
// Lattice highlight-mark plugin – converts ==text== to <mark> elements
export const MARK_RE = /==(?:[^\n]|(?:!))+?)==/g;

export function splitAtMarks(text: string): HastNode[] {
  const parts: HastNode[] = [];
  let lastIndex = 0;
  MARK_RE.lastIndex = 0;
  let match: RegExpExecArray | null;
  while ((match = MARK_RE.exec(text)) !== null) {
    if (match.index > lastIndex)
```

```

    parts.push({ type: 'text', value: text.slice(lastIndex, match.index) });
    parts.push({
      type: 'element', tagName: 'mark',
      properties: {},
      children: [{ type: 'text', value: match[1] }],
    });
    lastIndex = match.index + match[0].length;
  }
  if (lastIndex < text.length)
    parts.push({ type: 'text', value: text.slice(lastIndex) });
  return parts;
}

```

```

// Tauri command – read file with conflict detection
#[tauri::command]
pub async fn read_file(path: String) -> Result<FileReadResult, String> {
  let content = fs::read_to_string(&path)
    .map_err(|e| format!("Cannot read {path}: {e}"))?;
  let hash = compute_sha256(&content);
  Ok(FileReadResult { content, hash })
}

```

```

# Build a production release
export LATTICEBUILD_NO="MyBuild-001"
npm run tauri build

```

Blockquotes

Single-level blockquote. Great for callouts, citations, or side notes.

Nested blockquotes are also supported:

Inner quote — *"Documents as code, not documents as data."*

— Lattice design philosophy

GitHub-style Alerts (rendered by GitHub and Lattice)

[!NOTE] Useful information that users should know, even when skimming.

[!TIP] Optional advice for a better experience.

[!IMPORTANT] Key information users need to know to achieve their goal.

[!WARNING] Urgent information that needs immediate attention.

[!CAUTION] Advises about risks or negative outcomes.

Mathematics — KaTeX

Lattice renders mathematics via **KaTeX**. Use `$...$` for inline and `$$...$$` for display math.

Inline math

Einstein's famous equation $E = mc^2$ and the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ render inline.

The Euler identity $e^{i\pi} + 1 = 0$ is often called the most beautiful equation in mathematics.

Display math — Calculus

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$
$$\frac{d}{dx} \left[\int_a^{g(x)} f(t) dt \right] = f(g(x)) \cdot g'(x)$$

Display math — Linear Algebra

$$\mathbf{Ax} = \lambda \mathbf{x} \implies \det(\mathbf{A} - \lambda \mathbf{I}) = 0$$
$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

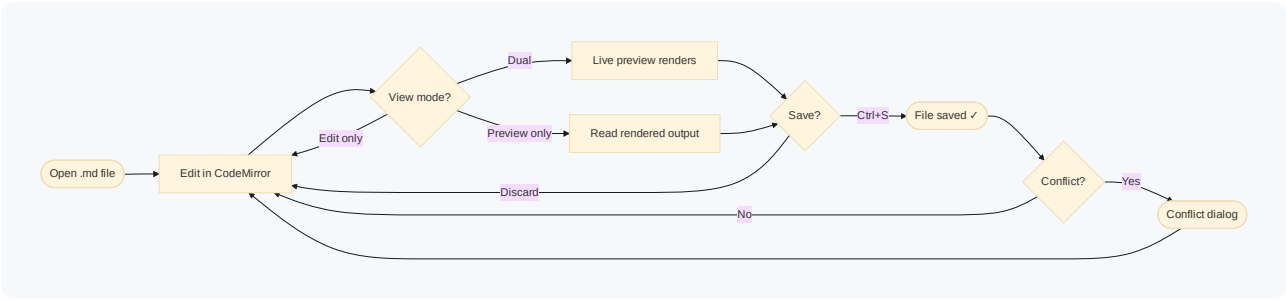
Display math — Series & Limits

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$
$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e$$

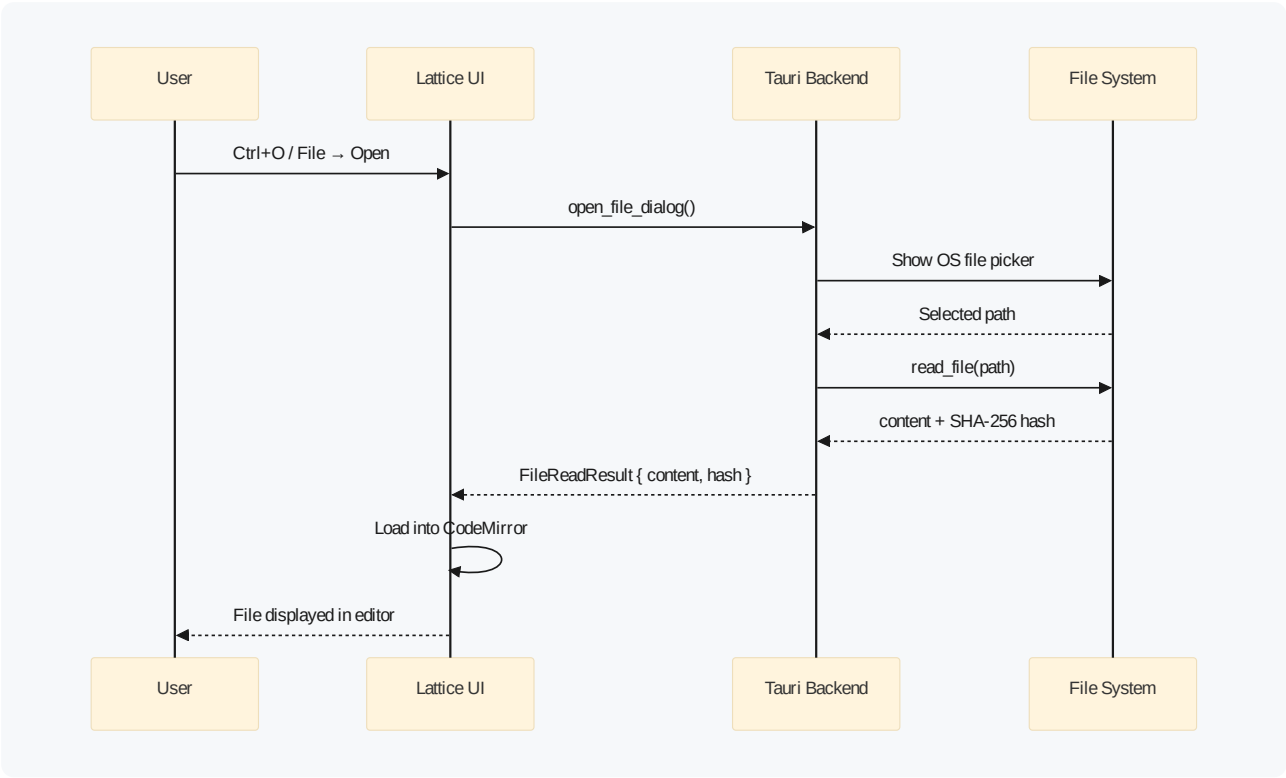
Mermaid Diagrams

Lattice renders ````mermaid```` fenced blocks inline in the preview.

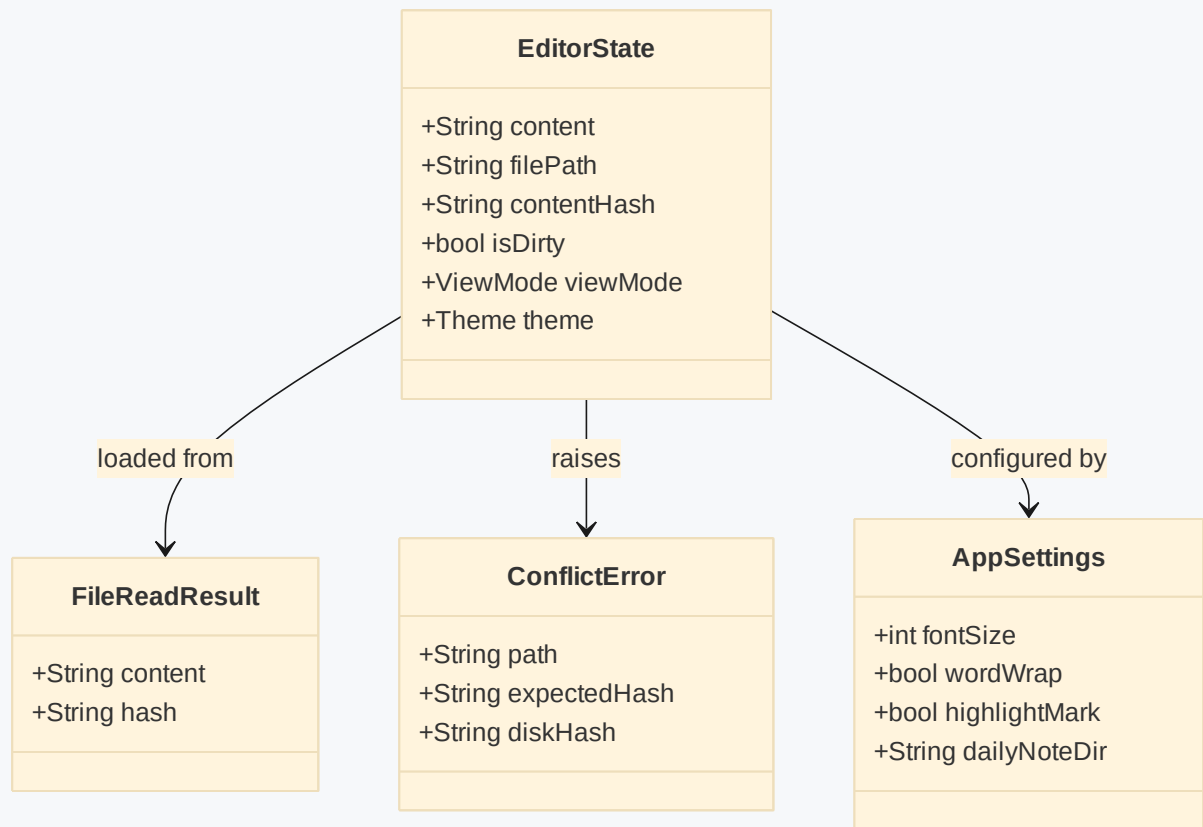
Flowchart — Lattice Edit Cycle



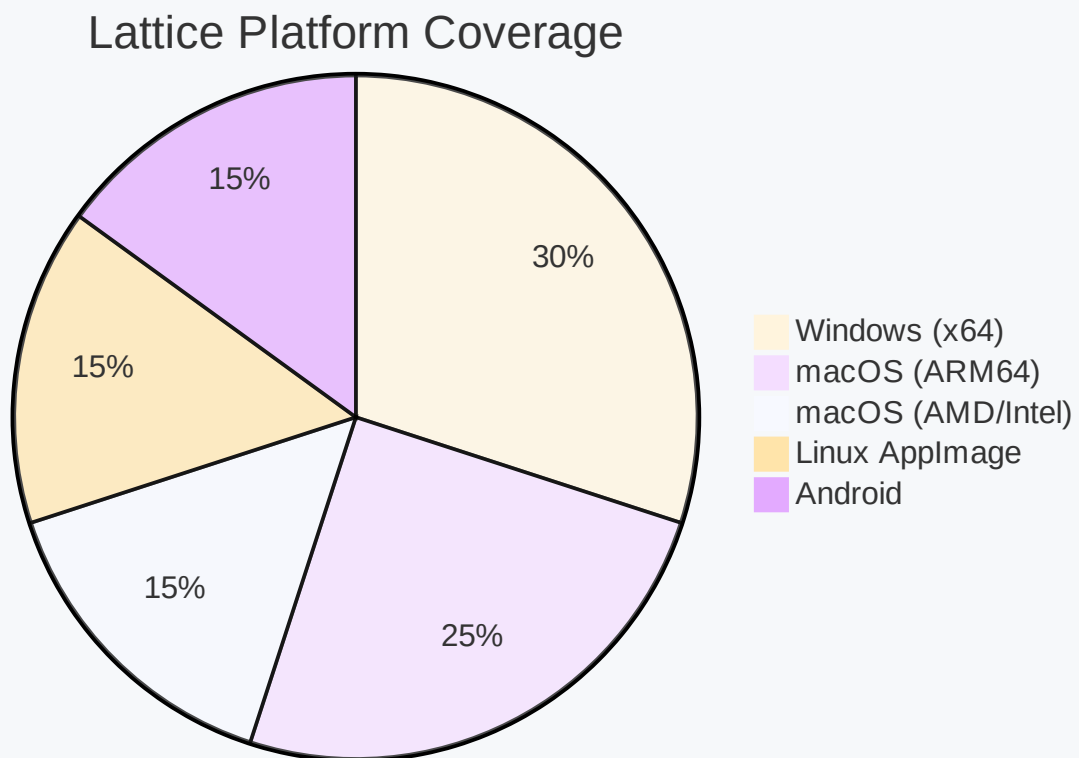
Sequence Diagram — File Open Flow



Class Diagram — Core Data Model



Pie Chart — Supported Platforms



The image file is automatically saved to a sibling folder named after the current document, and a Markdown reference is inserted at the cursor position:

! [Image] (demo_assets/img_1780575215039.png)

demo.md - lattice (v0.2.39 · windows-desktop-2026W23.0EEC)

XUSERPROFILE%_prj\lattice-ws\lattice\docs\demo\demo.md

207 ## Mathematics — KaTeX
208
209 Lattice renders mathematics via **KaTeX**. Use `\$....\$` for inline and
`\$\$...\$\$` for display math.
210
211 ### Inline math
212
213 Einstein's famous equation $E = mc^2$ and the quadratic formula
214 $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ render inline.
215
216 The Euler identity $e^{i\pi} + 1 = 0$ is often called the most
217 beautiful equation in mathematics.
218 ### Display math — Calculus
219
220 \$\$
221 \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}
222 \$\$
223
224 \$\$
225 \frac{d}{dx} \left(\int_a^{g(x)} f(t) dt \right) = f(g(x)) \cdot g'(x)
226 \$\$
227
228 ### Display math — Linear Algebra
229
230 \$\$
231 \mathbf{A} \mathbf{x} = \lambda \mathbf{x} \implies \det(\mathbf{A} - \lambda \mathbf{I}) = 0
232 \quad \text{or} \quad \det(\mathbf{A} - \lambda \mathbf{I}) = 0
233 \det(\mathbf{A} - \lambda \mathbf{I}) = 0
234 \$\$
235
236 \$\$
237 \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}
238 \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}
239 \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}
240 \$\$
241
242 ### Display math — Series & Limits
243
244 \$\$
245 e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots
246 = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots
247 \$\$
248
249 \$\$
250 \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e
251 \$\$
252 ---
253

Mathematics — KaTeX

Lattice renders mathematics via KaTeX. Use \$....\$ for inline and \$\$...\$\$ for display math.

Inline math

Einstein's famous equation $E = mc^2$ and the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ render inline.

The Euler identity $e^{i\pi} + 1 = 0$ is often called the most beautiful equation in mathematics.

Display math — Calculus

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$
$$\frac{d}{dx} \left[\int_a^{g(x)} f(t) dt \right] = f(g(x)) \cdot g'(x)$$

Display math — Linear Algebra

$$\mathbf{Ax} = \lambda \mathbf{x} \implies \det(\mathbf{A} - \lambda \mathbf{I}) = 0$$
$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Display math — Series & Limits

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$
$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e$$

Mermaid Diagrams

Lattice renders

```
graph LR
    A --> B
    B --> C
    C --> D
    D --> E
    E --> F
    F --> G
    G --> H
    H --> I
    I --> J
    J --> K
    K --> L
    L --> M
    M --> N
    N --> O
    O --> P
    P --> Q
    Q --> R
    R --> S
    S --> T
    T --> U
    U --> V
    V --> W
    W --> X
    X --> Y
    Y --> Z
    Z --> AA
    AA --> AB
    AB --> AC
    AC --> AD
    AD --> AE
    AE --> AF
    AF --> AG
    AG --> AH
    AH --> AI
    AI --> AJ
    AJ --> AK
    AK --> AL
    AL --> AM
    AM --> AN
    AN --> AO
    AO --> AP
    AP --> AQ
    AQ --> AR
    AR --> AS
    AS --> AT
    AT --> AU
    AU --> AV
    AV --> AW
    AW --> AX
    AX --> AY
    AY --> AZ
    AZ --> BA
    BA --> BB
    BB --> BC
    BC --> BD
    BD --> BE
    BE --> BF
    BF --> BG
    BG --> BH
    BH --> BI
    BI --> BJ
    BJ --> BK
    BK --> BL
    BL --> BM
    BM --> BN
    BN --> BO
    BO --> BP
    BP --> BQ
    BQ --> BR
    BR --> BS
    BS --> BT
    BT --> BU
    BU --> BV
    BV --> BW
    BW --> BX
    BX --> BY
    BY --> BZ
    BZ --> CA
    CA --> CB
    CB --> CC
    CC --> CD
    CD --> CE
    CE --> CF
    CF --> CG
    CG --> CH
    CH --> CI
    CI --> CJ
    CJ --> CK
    CK --> CL
    CL --> CM
    CM --> CN
    CN --> CO
    CO --> CP
    CP --> CQ
    CQ --> CR
    CR --> CS
    CS --> CT
    CT --> CU
    CU --> CV
    CV --> CW
    CW --> CX
    CX --> CY
    CY --> CZ
    CZ --> DA
    DA --> DB
    DB --> DC
    DC --> DD
    DD --> DE
    DE --> DF
    DF --> DG
    DG --> DH
    DH --> DI
    DI --> DJ
    DJ --> DK
    DK --> DL
    DL --> DM
    DM --> DN
    DN --> DO
    DO --> DP
    DP --> DQ
    DQ --> DR
    DR --> DS
    DS --> DT
    DT --> DU
    DU --> DV
    DV --> DW
    DW --> DX
    DX --> DY
    DY --> DZ
    DZ --> EA
    EA --> EB
    EB --> EC
    EC --> ED
    ED --> EE
    EE --> EF
    EF --> EG
    EG --> EH
    EH --> EI
    EI --> EJ
    EJ --> EK
    EK --> EL
    EL --> EM
    EM --> EN
    EN --> EO
    EO --> EP
    EP --> EQ
    EQ --> ER
    ER --> ES
    ES --> ET
    ET --> EU
    EU --> EV
    EV --> EW
    EW --> EX
    EX --> EY
    EY --> EZ
    EZ --> FA
    FA --> FB
    FB --> FC
    FC --> FD
    FD --> FE
    FE --> FF
    FF --> FG
    FG --> FH
    FH --> FI
    FI --> FJ
    FJ --> FK
    FK --> FL
    FL --> FM
    FM --> FN
    FN --> FO
    FO --> FP
    FP --> FQ
    FQ --> FR
    FR --> FS
    FS --> FT
    FT --> FU
    FU --> FV
    FV --> FW
    FW --> FX
    FX --> FY
    FY --> FZ
    FZ --> GA
    GA --> GB
    GB --> GC
    GC --> GD
    GD --> GE
    GE --> GF
    GF --> GG
    GG --> GH
    GH --> GI
    GI --> GJ
    GJ --> GK
    GK --> GL
    GL --> GM
    GM --> GN
    GN --> GO
    GO --> GP
    GP --> GQ
    GQ --> GR
    GR --> GS
    GS --> GT
    GT --> GU
    GU --> GV
    GV --> GW
    GW --> GX
    GX --> GY
    GY --> GZ
    GZ --> HA
    HA --> HB
    HB --> HC
    HC --> HD
    HD --> HE
    HE --> HF
    HF --> HG
    HG --> HH
    HH --> HI
    HI --> HJ
    HJ --> HK
    HK --> HL
    HL --> HM
    HM --> HN
    HN --> HO
    HO --> HP
    HP --> HQ
    HQ --> HR
    HR --> HS
    HS --> HT
    HT --> HU
    HU --> HV
    HV --> HW
    HW --> HX
    HX --> HY
    HY --> HZ
    HZ --> IA
    IA --> IB
    IB --> IC
    IC --> ID
    ID --> IE
    IE --> IF
    IF --> IG
    IG --> IH
    IH --> II
    II --> IJ
    IJ --> IK
    IK --> IL
    IL --> IM
    IM --> IN
    IN --> IO
    IO --> IP
    IP --> IQ
    IQ --> IR
    IR --> IS
    IS --> IT
    IT --> IU
    IU --> IV
    IV --> IW
    IW --> IX
    IX --> IY
    IY --> IZ
    IZ --> JA
    JA --> JB
    JB --> JC
    JC --> JD
    JD --> JE
    JE --> JF
    JF --> JG
    JG --> JH
    JH --> JI
    JI --> JJ
    JJ --> JK
    JK --> JL
    JL --> JM
    JM --> JN
    JN --> JO
    JO --> JP
    JP --> JQ
    JQ --> JR
    JR --> JS
    JS --> JT
    JT --> JU
    JU --> JV
    JV --> JW
    JW --> JX
    JX --> JY
    JY --> JZ
    JZ --> KA
    KA --> KB
    KB --> KC
    KC --> KD
    KD --> KE
    KE --> KF
    KF --> KG
    KG --> KH
    KH --> KI
    KI --> KJ
    KJ --> KK
    KK --> KL
    KL --> KM
    KM --> KN
    KN --> KO
    KO --> KP
    KP --> KQ
    KQ --> KR
    KR --> KS
    KS --> KT
    KT --> KU
    KU --> KV
    KV --> KW
    KW --> KX
    KX --> KY
    KY --> KZ
    KZ --> LA
    LA --> LB
    LB --> LC
    LC --> LD
    LD --> LE
    LE --> LF
    LF --> LG
    LG --> LH
    LH --> LI
    LI --> LJ
    LJ --> LK
    LK --> LL
    LL --> LM
    LM --> LN
    LN --> LO
    LO --> LP
    LP --&
```

Here is how it renders in `Dual (edit on the bottom) mode`:

The image shows the Lattice v0.2.39 application interface. At the top, a green bar displays the file path: %USERPROFILE%_prj\lattice-ws\lattice\docs\demo\demo.md. Below this is a dark navigation bar with a zoom slider set to 70%, a 'Dual (edit on bottom)' button, and a 'Menu' button. The main content area is titled 'Mathematics — KaTeX' and contains the following text: 'Lattice renders mathematics via KaTeX. Use $\$...$$ for inline and $\\$...\\$$ for display math.'

The 'Inline math' section shows Einstein's famous equation $E = mc^2$ and the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ rendered inline.

The 'Display math' section is titled 'Calculus' and shows the Euler identity $e^{i\pi} + 1 = 0$ is often called the most beautiful equation in mathematics. Below this, it shows the integral $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$ and the chain rule $\frac{d}{dx} \left[\int_a^{g(x)} f(t) dt \right] = f(g(x)) \cdot g'(x)$.

The 'Display math' section is titled 'Linear Algebra' and shows the equation $\mathbf{A}\mathbf{x} = \lambda\mathbf{x} \implies \det(\mathbf{A} - \lambda\mathbf{I}) = 0$ and the inverse of a 2x2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

The 'Display math' section is titled 'Series & Limits' and shows the Taylor series for e^x : $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$.

The bottom panel shows the source LaTeX code for the document, with line numbers 205 through 233. The code includes comments and LaTeX commands for rendering the equations shown above.

Tip: The assets folder name is derived from the document filename, keeping related

files grouped together and easy to version-control.

Links

- [Lattice on GitHub](#) — source code and issue tracker
- [Releases Page](#) — download installers
- [Release Site](#) — GitHub Pages release portal
- Relative link: [System Architecture Demo](#)
- Relative link: [Physics Equations Demo](#)

Auto-link: <https://github.com/blessia-blessini/lattice>

HTML in Markdown

Lattice renders a **safe whitelist** of inline HTML tags and passes everything else through as literal text. The linter flags all inline HTML — whitelisted tags get a **hint** (portability risk), non-whitelisted tags get a **warning** (not rendered here).

HTML element	Lattice preview	GitHub	Portable alternative
<sup>	✓ rendered	✓	KaTeX: x^2 → X^2 · Unicode: x^2
<sub>	✓ rendered	✓	KaTeX: H_2O → H_2O · Unicode: H_2O
<kbd>	✓ rendered	✓	Inline code: <code>ctrl+s</code>
 	✓ rendered	✓	Two trailing spaces or blank line
<mark>	⚠ plain text	✓	<code>==highlight==</code> (Lattice-native)
<details>	⚠ plain text	✓	Headings + TOC (no universal equivalent)
Other HTML	⚠ plain text	varies	Markdown equivalent (case-by-case)

Live examples of the whitelisted tags:

- Superscript: $X^2 + Y^3$
- Subscript: H_2O and CO_2

- Keyboard: press `Ctrl + S` to save

Keyboard Shortcuts Reference

Action	Windows / Linux	macOS
Save	<code>Ctrl+S</code>	<code>Cmd+S</code>
Open file	<code>Ctrl+O</code>	<code>Cmd+O</code>
New window	<code>Ctrl+N</code>	<code>Cmd+N</code>
Generate / refresh TOC	<code>Ctrl+Shift+T</code>	<code>Cmd+Shift+T</code>
Auto-align table	<code>Ctrl+Shift+L</code>	<code>Cmd+Shift+L</code>
Paste image from clipboard	<code>Ctrl+V</code>	<code>Cmd+V</code>
Print / Save as PDF	<code>Ctrl+P</code>	<code>Cmd+P</code>
Increase font size	<code>Ctrl++</code>	<code>Cmd++</code>
Decrease font size	<code>Ctrl+-</code>	<code>Cmd+-</code>
